

WASP-12b

R-0538

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_181206 · created 2026-07-21T23:13:43+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458458.82463 +/- 0.00056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.045
Transit depth (Rp/Rs)^2	2.21 +/- 1.33 [%]
Orbital Inclination (inc)	76.88 +/- 2.83
Airmass coefficient 1 (a1)	6.57 +/- 4.84
Airmass coefficient 2 (a2)	-1.7 +/- 0.93
Scatter in the residuals of the lightcurve fit is	77.18 %
Best Comparison Star	#1 - [233, 178]
Optimal Aperture	3.17
Optimal Annulus	9.4
Transit Duration (day)	0.106 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.045 vs 0.1178 (deviation 0.69σ)
Duration fit vs published	0.106 d vs 0.125 d (deviation 1.12σ)
Mid-time O–C	-0.2 min
Depth SNR	3.3
Residual scatter	77.18 %

Light curve

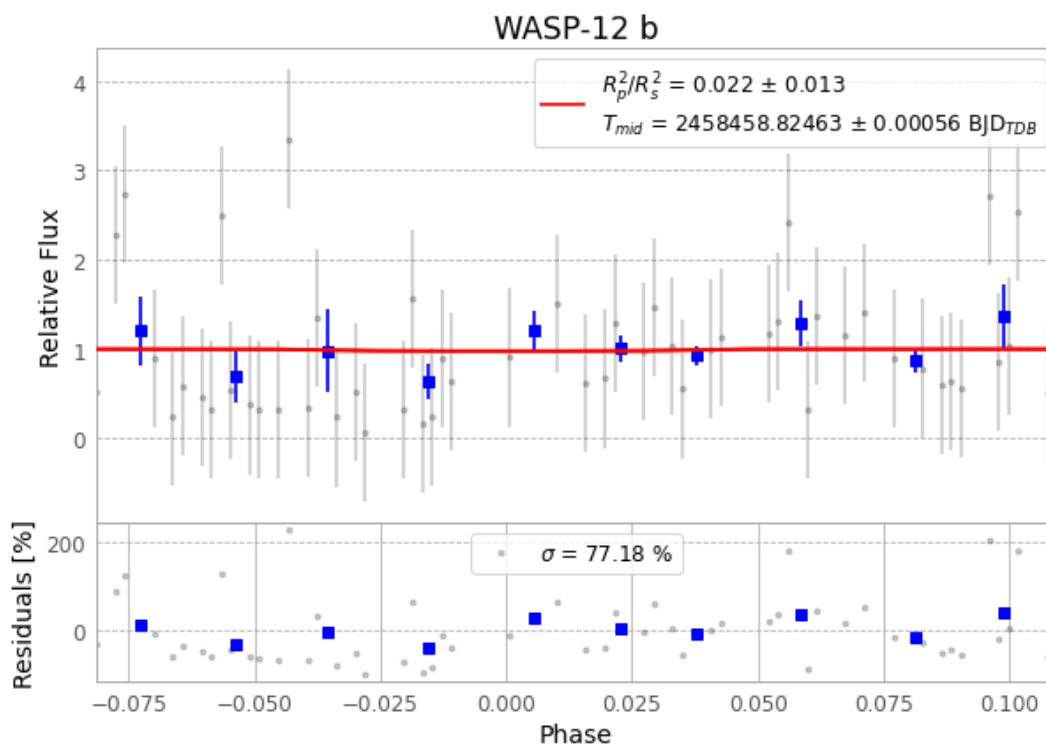


Figure 1 — FinalLightCurve_WASP-12 b_2018-12-05.png (SHA-256 2ca64b88e35b976b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [345, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 355cd48accb4ae61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

101 FITS frames (66 MB), WASP-12181206053012.FITS → WASP-12181206103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181206.log (e3aff97764ed...), FinalLightCurve_WASP-12 b_2018-12-05.png (2ca64b88e35b...), FinalLightCurve_WASP-12 b_2018-12-05.csv (48731b112e24...)

Compute resources

Wall time 315.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 23:13Z.